One-Sided Littlewood Caps, Quantitative Word Products, and Physical Spectral Certificates

A scoped Ram-calculus study of asymptotically minimal maxima

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Abstract

The September 23 source claims asymptotically minimal maximum modulus for real Littlewood polynomials at every integer length. Its one-sided conclusion does not control the minimum modulus, and its Morse application does not identify the entire mean-zero Koopman spectrum. We prove a source-specific extension: summable one-sided squared-modulus caps on mixed-radix binary blocks give a bounded density positive almost everywhere, with explicit total-variation, finite-L^p, and relative lower-tail certificates. Exact word normalization replaces an independence assumption. A physical correlation argument identifies this density on the Morse coordinate cyclic subspace, conditional on the declared classical realization inputs. We also give finite autocorrelation, grid-to-circle, and mass-scaled Ram Gram certificates, an explicit auxiliary-recursion rate, and worked rational examples. Original Ram frame proofs are used with their finite-model and perturbation margins; several local FFT normalization errors in an original comparison input are corrected. No effective rate for the global sign construction, historical novelty, or full simple-Lebesgue Koopman theorem is asserted. Contribution assessment: C3/5, the proved one-sided product limit and quantitative spectral-window certificates give a notable local strengthening while the global construction and full-space cyclicity remain separate obligations.

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1 Normalizations, claims, and evidence boundaries

All circle integrals use probability Haar measure dt on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and $\mathrm{e}(t) = \exp(2\pi i t)$. For a real binary word $a = (a_0, \dots, a_{N-1})$, put

$$P_a(z) = \sum_{j=0}^{N-1} a_j z^j, \qquad \rho_a(t) = \frac{|P_a(\mathrm{e}(t))|^2}{N}, \qquad C_u(a) = \sum_{j=0}^{N-1-u} a_j a_{j+u}. \tag{1.1}$$

Set $c_0 = 1$, $c_{\pm u} = C_u/N$ for $1 \le u < N$, and $c_u = 0$ for $|u| \ge N$. Then

$$\rho_a(t) = \sum_{u\in\mathbb{Z}} c_u \mathrm{e}(ut), \quad \int \rho_a = 1, \quad \int (\rho_a - 1)^2 = \frac{2}{N^2}\sum_{u=1}^{N-1} C_u^2 = F(a)^{-1}. \tag{1.2}$$

For $N \ge 2$, $C_{N-1} = \pm 1$, so the merit factor $F(a)$ is finite. We use $F^{-1} = 0$ for the length-one word. There is no angular 2π factor in these integrals.

The cached source [1] asserts

$$m_N := \min_{a\in\{-1,1\}^N} \frac{\|P_a\|_\infty}{\sqrt{N}} \longrightarrow 1$$

through all integers, with the word chosen separately at each length. This is a source claim, not a new theorem of this report. In particular, the source already proves the sharp elementary consequence

$$F(a)^{-1} \le \|\rho_a\|_\infty - 1, \qquad \max_{|a|=N} F(a) \ge (m_N^2 - 1)^{-1}. \tag{1.3}$$

Indeed $\rho_a^2 \le \|\rho_a\|_\infty \rho_a$, and (1.2) proves the assertion. Neither this identity nor the source's concentration counterexamples are counted as local innovations.

All ten TeX modules, the bibliography, and the source README were read literally. That is complete reading of this cached manuscript, not an independent proof of every external theorem it invokes. In particular, hypergraph edge coloring, arbitrary-vector partial coloring, and classical Morse realization facts are imported. The conditional consequences below name the precise inputs they use. Other Littlewood and three-torus companion manuscripts are not used as premises.

1.1 Current public comparison

The primary-literature refresh is dated October 8, 2026. Balister, Bollobás, Morris, Sahasrabudhe, and Tiba [2] prove constant-factor two-sided flatness. That published result neither supplies a cap tending to one nor follows from ignoring deep zeros. Erdélyi's August 2026 preprint [3] states, in our length convention,

$$\|P_a\|_\infty^2 \geq N + \frac{(N-1)^{1/3}}{38}.$$

Its relative squared excess tends to zero, so the stated lower bound is compatible with the source's claimed asymptotic. We cite the statement, not an independent verification of its entire proof. Erdélyi's 2025 fixed-prefix obstruction [4] concerns all partial sums of one unimodular power series; it is not an obstruction to choosing signs independently at different lengths. Our mixed-radix prefixes below are not asserted to become uniformly two-sided flat. A public repository listing or formalization metadata is not itself a proof certification.

2 The source mechanism and its quantifiers

Module	Mathematical role and retained condition
Introduction; reduction	Reduce to real $X_{N,k} \in [-1,1]$ with nearly full mean-square mass. Fix the accuracy parameter before sending N to infinity.
Packing	Pack signed closed intervals for pairwise nonparallel integer forms. Pair ranks, not full independence of all forms, control codegrees. The large packing prime is auxiliary and imposes no divisibility on N.
Spreading	Generate near-unit energy, tune finitely many nonzero quadratic curvatures, and truncate with a summable off-stationary Fourier tail. The uniform stationary-phase error alone cannot be summed over an infinite frequency lattice.
Sampling	Use disjoint physical blocks and packed spectral intervals. At most one stationary leading term survives at each angle. Equal curvatures still require a nonintegral center difference.
Rounding	Partial coloring at dyadic defect scales, with total defect nonincreasing. The final grid-to-circle estimate applies to the rounding error.
Flatness; merit–Morse	Finite-exponent modulus flatness and the already-known merit estimate; then a Morse coordinate measure with L^2 density via classical inputs.
Concentration	Correct an infimum/supremum reversal and an invalid unrestricted weighted claim. These are source results retained here, not new contributions.

2.1 A usable but limited auxiliary rate

The source's real torus recursion is

$$p_0 = 0, \qquad p_j = p_{j-1} + \tfrac{1}{2}(1 - p_{j-1}^2)\cos(2\pi y_j).$$

It has $|p_j| \leq 1$, mean zero, and $M_j = \int p_j^2$ satisfies $M_j - M_{j-1} = \frac{1}{8}\int (1-p_{j-1}^2)^2$.

Proposition 2.1. *For every integer $j \geq 0$,*

$$1 - M_j \leq \frac{8}{j+8}.$$

Thus $d \ge \max(2, \lceil 8/\delta - 8 \rceil)$ is sufficient for $M_d \ge 1 - \delta$.

Proof. Writing $u_j = 1 - M_j$, Jensen gives $0 \le u_{j+1} \le u_j(1 - u_j/8)$. If a term is zero there is nothing left to prove. Otherwise

$$\frac{1}{u_{j+1}} \ge \frac{1}{u_j(1 - u_j/8)} \ge \frac{1}{u_j} + \frac{1}{8}.$$

Start with $u_0 = 1$ and sum. □

This is an explicit local dimension budget, not a useful bound for the final length. The subsequent Fourier support, small curvatures, cutoff derivatives, packing, and partial coloring still carry substantial costs.

2.2 Finite energy is a Gram statement

For the source's fixed blocks and modes, write the sampled coefficient as

$$X_k = \sum_{h,a} c_a \chi_h(k/N) \mathrm{e}\Big(k\, a \cdot \theta_h + \tfrac{N}{2} \lambda_a (k/N - x_h^*)^2 \Big).$$

Conjugate symmetry makes X_k real. Its energy is exactly a quadratic form in the coefficients c_a; the within-block Gram entry for a, a' is

$$\frac{1}{N} \sum_k \chi_h(k/N)^2 \mathrm{e}\Big(k\beta + \tfrac{N}{2} \kappa (k/N - x_h^*)^2 \Big), \quad \beta = (a - a') \cdot \theta_h, \quad \kappa = \lambda_a - \lambda_{a'}.$$

Cross-block entries vanish. With all auxiliary data fixed, off-diagonal entries are $O(N^{-1/2})$ when $\kappa \neq 0$, and $O(N^{-1})$ when $\kappa = 0,\ \beta \notin \mathbb{Z}$. The latter exclusion follows from packed centers, not from unequal curvatures. Pair errors can cancel; taking absolute values is a sufficient bound, not an identity.

The reduction uses

$$K_\delta = \sqrt{\frac{(1+\delta)^3}{1-\delta}}, \qquad \liminf_N \frac{1}{N} \sum_k X_k^2 \ge 1 - 7\delta$$

and the source's rounding estimate with $\mu = \frac{1}{2} \sum_k (1 - |X_k|)$. It gives $\limsup_N m_N \le K_\delta + 2C\sqrt{\delta \log(20/\delta)}$, then sends $\delta \downarrow 0$. Replacing that order by a chosen $\delta(N)$ requires uniform constants which the manuscript does not provide.

3 One-sided caps and quantitative product limits

Lemma 3.1 (Mean and cap)**.** *On a probability space, suppose $0 \le q \le 1 + \delta$, $\int q = 1$, and $\delta \ge 0$. Then*

$$\|q - 1\|_1 \le \frac{2\delta}{1+\delta}, \tag{3.1}$$

$$\int |q - 1|^p \le \frac{\delta + \delta^p}{1+\delta} \quad (p \ge 1), \tag{3.2}$$

$$\mu\{q \le a\} \le \frac{\delta}{1 + \delta - a} \quad (0 \le a < 1). \tag{3.3}$$

Proof. The positive and negative parts of $q-1$ have equal integrals, and $(q-1)_+ \le \delta q/(1+\delta)$, proving (3.1). The convex function $|x-1|^p$ lies below its chord on $[0,1+\delta]$; integration of that chord proves (3.2). If $u = \mu\{q \le a\}$, then $1 \le au + (1+\delta)(1-u)$, proving (3.3). □

These classical inequalities are sharp for general probability densities. Their extremizers need not be binary-polynomial densities. They control the measure of deep dips, not their absence or a reciprocal density norm.

Theorem 3.2 (Summable one-sided product certificate)**.** *Let q_j be measurable functions on a probability space, positive almost everywhere, with $\int q_j = 1$ and*

$$0 \le q_j \le 1+\delta_j, \qquad \delta_j \ge 0, \qquad \sum_{j=1}^{\infty} \delta_j < \infty.$$

Assume separately that every product $w_n = \prod_{j=1}^{n} q_j$ has integral one. Set

$$B = \prod_{j\ge 1}(1+\delta_j), \quad Q_n = \prod_{j>n}(1+\delta_j), \quad \tau_n = \sum_{j>n} \frac{\delta_j}{1+\delta_j}.$$

Then w_n converges almost everywhere and in every finite L^p, $1 \le p < \infty$, to a density w with $0 < w \le B$ almost everywhere and $\int w = 1$. Moreover,

$$\|w - w_n\|_1 \le 2(1 - Q_n^{-1}), \tag{3.4}$$

$$\|w - w_n\|_p^p \le 2B^{p-1}(1 - Q_n^{-1}), \tag{3.5}$$

$$\mu\{w < e^{-2s} w_n\} \le \min(1, 2\tau_n/s), \quad 0 < s \le \tfrac{1}{2}. \tag{3.6}$$

The L^1 bound is twice the usual total-variation distance. No independence of the factors is assumed.

Proof. Lemma 3.1 and Tonelli imply $\sum_j |q_j - 1| < \infty$ almost everywhere. Off the countable union of the zero sets, all factors are positive. Eventually $|q_j - 1| \le 1/2$, and then $|\log q_j| \le 2|q_j - 1|$. The tail logarithms are absolutely summable; the finite initial product is positive. This proves pointwise convergence to a finite positive limit. Since $w_n \le B$, dominated convergence gives L^p convergence and preserves the integral.

For $m > n$, let $Q_{n,m} = \prod_{n<j\le m}(1+\delta_j)$. We have $w_m \le Q_{n,m} w_n$, so $(w_m - w_n)_+ \le (1 - Q_{n,m}^{-1}) w_m$. Both products have mass one; hence $\|w_m - w_n\|_1 \le 2(1 - Q_{n,m}^{-1})$. Pass to the limit. Since $w, w_n \in [0, B]$, $|w - w_n|^p \le B^{p-1}|w - w_n|$, proving (3.5).

Finally $D_n = \sum_{j>n} |q_j - 1|$ has integral at most $2\tau_n$. Outside a set of measure at most $2\tau_n/s$, $D_n \le s \le 1/2$. There every tail factor lies within $1/2$ of one and $\sum_{j>n} \log q_j \ge -2D_n \ge -2s$. Exponentiate. □

Remark 3.3. The product-mass hypothesis is essential. Two individually normalized equal factors $q_1 = q_2 = 1 + \varepsilon\cos(2\pi t)$ have product integral $1 + \varepsilon^2/2$, not one. Thus a generic moment bank cannot assume mass preservation merely because its individual atoms are normalized. Conversely, the theorem does not require globally integrable $|\log q_j|$, two-sided uniform flatness, or a positive essential infimum. Its bounded-product domination supplies uniform integrability explicitly.

4 Binary products and the physical Morse coordinate

For words A, B of lengths a, b, define the source's ordered product $(A \times B)_{s+at} = A_s B_t$, $0 \le s < a,\ 0 \le t < b$. The polynomial identity

$$P_{A\times B}(z) = P_A(z) P_B(z^a) \tag{4.1}$$

follows by expanding the two sums. Every exponent $s + at$ has exactly one representation in the indicated ranges, so the coefficients remain binary and there are no gaps.

Corollary 4.1 (No-floor word certificate)**.** *Let B_j be binary words of lengths $n_j \ge 2$, and suppose*

$$\left\| P_{B_j} \right\|_\infty^2 / n_j \le 1 + \delta_j, \qquad \sum_j \delta_j < \infty.$$

Put $h_0 = 1,\ h_j = n_1 \cdots n_j$, $A_m = B_1 \times \cdots \times B_m$, and $q_j(t) = \rho_{B_j}(h_{j-1} t)$. Then

$$w_m(t) = \rho_{A_m}(t) = \prod_{j \le m} q_j(t)$$

satisfies every conclusion of Theorem 3.2. In particular its limit $w\,dt$ is equivalent to Haar measure and has bounded density. The block polynomials may have unit-circle zeros.

Proof. Integer dilation preserves Haar measure, so each q_j has integral one. A nonzero polynomial has only finitely many circle zeros; a fixed integer dilation preserves nullity of that zero set. Identity (4.1) and Parseval give $\int w_m = 1$ exactly. Apply Theorem 3.2. □

There is no probabilistic independence or unjustified infinite-product normalization here. This is the specific replacement for a two-sided uniform-flatness or external uniform-integrability assumption.

4.1 Realization gates and correlation identification

We use precisely the following classical Morse inputs, as stated in Definition 3 and Facts 1–2 of Downarowicz–Lacroix [5]: words beginning with 1 define a binary Morse system; if both letter frequencies stay away from zero, its invariant measure is unique; the resulting binary system has simple spectrum. These realization and simplicity facts are imported, not proved by a finite Gram calculation. Their Theorem 2 is the source's existing route from large merit factors to an L^2 coordinate density. Our following argument uses the stronger cap hypotheses instead.

Theorem 4.2 (Physical coordinate limit)**.** *Assume the hypotheses of Corollary 4.1, $B_j(0) = 1$, that every block contains both signs, and $n_j \longrightarrow \infty$. For the Morse system supplied by the declared classical inputs, let μ be its invariant probability measure, $Sx(k) = x(k+1)$, and $f(x) = x(0)$. Then $\int f\,d\mu = 0,\ \|f\|_2 = 1$, and the scalar spectral measure defined by*

$$\int_{\mathbb{T}} \mathrm{e}(ut)\, d\sigma_f(t) = \int f(x) f(S^u x)\, d\mu(x)$$

is exactly $w(t)\,dt$. Consequently the restriction of $Ug = g \circ S$ to $\mathcal{H}_f = \overline{\mathrm{span}}\{U^u f : u \in \mathbb{Z}\}$ has simple Lebesgue type, with the explicit density-tail bounds in Theorem 3.2. This does not assert $\mathcal{H}_f = L^2_0(\mu)$.

Proof. At $z = 1$ the block imbalance is bounded by

$$\frac{|\sum B_j(k)|}{n_j} \le \sqrt{\frac{1+\delta_j}{n_j}} \longrightarrow 0.$$

The two letter frequencies therefore tend to $1/2$. Since every initial block contains both signs, their frequencies are bounded away from zero uniformly over all blocks. Under these classical realization hypotheses unique ergodicity identifies the orbit averages of the continuous function $f\,U^u f$ with its integral. The finite prefix A_m, of length h_m, differs from the corresponding full orbit correlation average by at most $|u|/h_m$, since only the pairs crossing its endpoint are omitted. Thus for each fixed integer $u \ge 0$,

$$\int f\,U^u f\,d\mu = \lim_m \frac{C_u(A_m)}{h_m} = \lim_m \int_{\mathbb{T}} \mathrm{e}(ut) w_m(t)\,dt = \int_{\mathbb{T}} \mathrm{e}(ut) w(t)\,dt.$$

The negative coefficients agree by symmetry, so uniqueness of Fourier coefficients of finite measures identifies $\sigma_f = w\,dt$. Also $|P_{A_m}(1)|/h_m \le \sqrt{B/h_m} \to 0$; the same orbit-average argument gives $\int f = 0$. Binary values give $\|f\|_2 = 1$.

The map $\sum a_u U^u f \mapsto \sum a_u \mathrm{e}(ut)$ is an isometry into $L^2(w\,dt)$ by the correlation identity. Trigonometric polynomials are dense there: a vector orthogonal to them defines a finite measure with every Fourier coefficient zero, and hence is zero. Completion identifies the entire cyclic space with $L^2(w\,dt)$. Multiplication by $\sqrt{w}$ is a unitary map from this space onto $L^2(dt)$, since $w > 0$ almost everywhere. It commutes with multiplication by $\mathrm{e}(t)$, proving the spectral assertion. □

Conditional on the source's all-length cap theorem, one can choose nonconstant blocks with $n_j \ge j + 2$ and any prescribed positive summable cap budget, then multiply each word by its first sign. This changes no modulus and gives $B_j(0) = 1$. Thus the stronger conclusion of Theorem 4.2 is available from that source input. No effective bound on the required n_j has been supplied. The classical fact of simple full spectrum does not identify its maximal type with the coordinate type: additional singular or atomic components may be supported on $\mathcal{H}_f^{\perp}$. The full-space Banach question requires that complement to vanish.

4.2 Tail schedules and their limitations

For example, if $\delta_j \le \eta 2^{-j}$, then

$$B \le e^{\eta}, \qquad Q_n \le e^{\eta 2^{-n}}, \qquad \|w - w_n\|_1 \le 2\eta 2^{-n}.$$

Therefore each correlation is approximated with absolute error at most $2\eta 2^{-n}$, and $\|w - w_n\|_p^p \le 2e^{(p-1)\eta}\eta 2^{-n}$. These are rates in the *chosen block index*, not rates in the polynomial length h_n. They allow zeros in every factor and do not invoke a fixed positive floor.

The cap and mean alone cannot control a logarithmic norm. On a probability space with a set of measure $0 < \epsilon < \delta/(1+\delta)$, take $q = e^{-K/\epsilon}$ there and choose its constant value on the complement to make $\int q = 1$. That second value is below $(1-\epsilon)^{-1} < 1+\delta$, while $\int(-\log q)_+ \ge K$. Thus arbitrarily large logarithmic cost is consistent with an arbitrarily small fixed cap excess. This example concerns general densities, not a constructed binary polynomial. The product proof uses summable absolute deviations and almost-everywhere positivity instead of such a nonexistent uniform logarithmic bound.

5 Finite modulus and autocorrelation certificates

Proposition 5.1 (A posteriori whole-circle cap)**.** *Let a be a specified binary word of length N, let $M \geq 1$, and let g be a rigorous upper bound for all values $\rho_a(\ell/M)$, $0 \leq \ell < M$. Define*

$$S_2(a) = \sum_{u=1}^{N-1} u^2 |C_u(a)|, \qquad U = g + \frac{\pi^2 S_2(a)}{NM^2}.$$

Then $\|\rho_a\|_\infty \leq U$. In a rational certificate one may replace π by any verified rational upper bound. Together with exact autocorrelations this gives the consistency gate

$$1 + \frac{2}{N^2} \sum_{u=1}^{N-1} C_u^2 \leq \|\rho_a\|_\infty \leq U. \tag{5.1}$$

Proof. The real Fourier expansion in (1.2) gives $\|\rho_a''\|_\infty \leq 8\pi^2 S_2(a)/N$. At a maximum t_*, $\rho_a'(t_*) = 0$. A nearest grid point is at circular distance at most $1/(2M)$. Taylor's theorem with the second-derivative bound gives the asserted excess at t_*. The lower bound follows from $\int \rho_a^2 \leq \|\rho_a\|_\infty \int \rho_a$. □

This is a certificate for one supplied word, not a search algorithm or an asymptotic existence proof. Unlike a first-derivative estimate, its mesh loss is $O(M^{-2})$ with an explicit autocorrelation constant. The actual dependence of S_2 on the word is retained.

There is a completely rational way to enclose the grid data. Reduce $u\ell/M$ modulo one to $r \in [-1/2, 1/2]$. For a rational π_0 with $|\pi - \pi_0| \leq \varepsilon_\pi$, Taylor's theorem gives

$$\left| \cos(2\pi r) - \sum_{k=0}^{d} \frac{(-1)^k (2\pi_0 r)^{2k}}{(2k)!} \right| \leq 2|r|\varepsilon_\pi + \frac{|2\pi_0 r|^{2d+2}}{(2d+2)!}. \tag{5.2}$$

Use Taylor degree $2d+1$, whose odd coefficient is zero, and the global derivative bound one for cosine. Multiply each enclosure by $2C_u/N$, retaining its sign, and add one. No agreement between two floating-point samples is used as a proof.

5.1 Toeplitz windows and their exact mass

For $L \geq 1$, let $T_L(a) = (c_{j-i})_{0 \leq i,j < L}$. For a coefficient column $x \in \mathbb{C}^L$, direct integration gives

$$x^* T_L(a) x = \int_{\mathbb{T}} \left| \sum_{j=0}^{L-1} x_j \mathrm{e}(jt) \right|^2 \rho_a(t)\, dt. \tag{5.3}$$

In particular $T_L(a)$ is positive definite: $\rho_a > 0$ almost everywhere and a nonzero trigonometric polynomial cannot vanish almost everywhere. Its entries are exact rational numbers.

Proposition 5.2 (Three sufficient finite margins)**.** *If $\rho_a \leq 1 + \delta$, then $T_L(a) \preceq (1+\delta) I$ and*

$$T_L(a) \succeq a_L I, \qquad a_L = \max\{1 - (L-1)\delta,\ 1 - \sqrt{D_L},\ 1 - G_L\},$$

where

$$D_L = 2 \sum_{u=1}^{\min(L-1, N-1)} (L-u)(C_u/N)^2, \qquad G_L = \max_{0 \leq i < L} \sum_{\substack{0 \leq j < L \\ j \neq i}} |c_{j-i}|.$$

Also $D_L \leq L/F(a)$. A nonpositive sufficient bound means inconclusive, not singular.

Proof. Equation (5.3) proves the upper bound. Since $\operatorname{tr} T_L = L$, the other $L-1$ eigenvalues contribute at most $(L-1)(1+\delta)$, giving the first lower bound. The exact squared Frobenius norm of $T_L - I$ is D_L; the operator norm is no larger. The maximum absolute row sum bounds the same operator norm by G_L. Finally bound every $L-u$ by L and use (1.2). □

These elementary matrix tools become useful certificates only because the entries, angular normalization, finite window, and cap remainder have been specified. For orbit exponents $-K, \dots, K$, one needs correlations through $|u| \le 2K$, not merely K.

Corollary 5.3 (Prefix-to-limit frame margin)**.** *For the density w in Theorem 3.2, let $T_L(w)_{ij} = \int \mathrm{e}((j-i)t)w(t)\,dt$. With $\epsilon_n = 2(1-Q_n^{-1})$,*

$$\|T_L(w) - T_L(w_n)\|_{\mathrm{op}} \le (L-1)\epsilon_n.$$

Hence a certified rational inequality $T_L(w_n) \succeq (a + (L-1)\epsilon_n)I$, $a > 0$, gives $T_L(w) \succeq aI$. In particular $L_n\epsilon_n \to 0$ is a sufficient moving-window transfer budget, not by itself a uniform positive frame bound.

Proof. Each off-diagonal entry differs by at most $\|w - w_n\|_1 \le \epsilon_n$. Both measures have mass one, so the diagonal differences vanish exactly. The absolute row-sum bound is $(L-1)\epsilon_n$; apply the variational characterization of the least eigenvalue. □

6 Original Ram proofs and physical finite banks

The original TR248 [7], not just its summary in the private spectral comparison [6], was inspected at its finite-model, perturbation, physical-mass, and exact moment-certificate proofs. Its central distinction is retained: a finite invertible matrix controls the named finite model, not its infinite-dimensional orthogonal complement. The spectral comparison's product and moving-window proofs were also read in full. The present no-floor theorem differs from its logarithmic exceptional-set criterion: here exact word mass and summable one-sided caps directly supply both convergence and uniform integrability, with explicit tails.

6.1 A weighted Gram certificate, not a coordinate renaming

Consider finite exponential/moment combinations

$$v_\ell(t) = \sum_{j=0}^{L-1} R_{j\ell}\mathrm{e}(jt), \qquad R \in \mathbb{C}^{L\times d}, \quad \operatorname{rank} R = d.$$

Such a bank is a finite Ram spectral model: its coefficients may encode modulations, weights, or finite moment combinations, but its physical space must still be declared. Its Haar mass and density-weighted mass are

$$M = R^*R \succ 0, \qquad G_a = R^*T_L(a)R, \qquad G_w = R^*T_L(w)R. \tag{6.1}$$

If $aI \preceq T_L(a) \preceq bI$, then $aM \preceq G_a \preceq bM$, by testing Ry for each y. Similarly Corollary 5.3 gives

$$-(L-1)\epsilon_n M \preceq G_w - G_{w_n} \preceq (L-1)\epsilon_n M.$$

A large eigenvalue of the raw matrix G_a is not a physical margin unless M is accounted for. For rational R, exact positive pivots in an LDL^* factorization of $G_a - aM$ prove a strict certificate; zero pivots require a valid semidefinite treatment rather than division by zero.

Here is the original perturbation argument in the form needed for this application. If a finite physical synthesis map V satisfies $AI \preceq V^*V \preceq BI$, and its implementation is $\widetilde{V} = V + E$, $\|E\| \le \eta < \sqrt{A}$, then

$$(\sqrt{A} - \eta)^2 I \preceq \widetilde{V}^*\widetilde{V} \preceq (\sqrt{B} + \eta)^2 I. \tag{6.2}$$

Indeed $\left\|\widetilde{V}x\right\| \ge \|Vx\| - \|Ex\|$ and the ordinary triangle inequality gives the upper bound. For observations $y = Vx + e$, the canonical left inverse of $\widetilde{V}$ has error at most $(\|e\| + \eta\,\|x\|)/(\sqrt{A} - \eta)$. This follows from $\widetilde{V}^\dagger y - x = \widetilde{V}^\dagger(e - Ex)$. These are imported classical frame tools with a new explicit source of A, not new general frame theorems.

6.2 The remaining full-space gate

For clarity we restate and prove the relevant part of the Ram moving-window criterion. All maps must take values in the *same* physical Hilbert space $\mathcal{H} = L^2_0(\mu)$.

Proposition 6.1 (Conditioned cyclicity transfer)**.** *Let U be an actual Koopman operator and $f \in \mathcal{H}$. For $K_J \to \infty$ define $S_J x = \sum_{|u| \le K_J} x_u U^u f$. Let $T_J : \mathbb{C}^{2K_J+1} \to \mathcal{H}$ satisfy $T_J^* T_J \succeq a_J I$, $a_J > 0$. Let increasing finite sets $\mathcal{D}_J$ of unit vectors have dense union in the unit sphere of $\mathcal{H}$. Suppose*

$$\max_{\psi \in \mathcal{D}_J} \operatorname{dist}(\psi, \operatorname{ran} T_J) \le \zeta_J \to 0, \qquad \frac{\|T_J - S_J\|}{\sqrt{a_J}} \to 0.$$

Then $\overline{\operatorname{span}}\{U^u f : u \in \mathbb{Z}\} = \mathcal{H}$.

Proof. Project a unit vector $\psi \in \mathcal{D}_J$ orthogonally onto $\operatorname{ran} T_J$, writing the projection $T_J x$. Its norm is at most one, so $\|x\| \le a_J^{-1/2}$. Consequently

$$\operatorname{dist}(\psi, \mathcal{H}_f) \le \zeta_J + \|(T_J - S_J)x\| \le \zeta_J + \|T_J - S_J\| / \sqrt{a_J}.$$

For each vector in the increasing union let J tend to infinity; then use density and closedness. □

If each column is within ξ_J of its actual Koopman counterpart, then $\|T_J - S_J\| \le \sqrt{2K_J + 1}\,\xi_J$, by Cauchy–Schwarz. Fixed-column convergence without this moving-window budget is insufficient. Neither the source nor the tests below establish the density premise for the Morse coordinate on all L^2_0. The independent conditions $U1 = 1$, positivity, multiplicativity on bounded functions, and a common invariant probability measure cannot be replaced by an arbitrary unitary matrix.

6.3 Corrections in the original spectral-computation input

Fresh reading of TR9 [8] found localized errors that must not propagate into an FFT-based certificate. With its convention $W_N = e^{-2\pi i/N}$,

$$W_{N/2}^{-\nu N/2} = 1,$$

not $(-1)^\nu$. Thus its unnormalised half-sums are exactly $N/2$-periodic. The two butterfly outputs are

$$g[x] = \frac{\mathcal{E}[x] + W_N^{-x}\mathcal{O}[x]}{N}, \quad g[x + N/2] = \frac{\mathcal{E}[x] - W_N^{-x}\mathcal{O}[x]}{N}.$$

The factor $1/2$ is correct only after normalizing the half-transforms by $2/N$, which the displayed definitions there do not do. A Taylor truncation indexed $0, \dots, d$ uses $d + 1$ separable terms, of rank *at most* $d + 1$, and generally $d + 1$ FFTs. An arbitrary explicitly supplied dense kernel has a quadratic input-reading cost in the worst case; this does not prove the claimed exact lower

counts of both N^2 multiplications and N^2 additions. The valid retained conclusion is that an output-dependent arbitrary kernel lacks the standard butterfly reuse, whereas a supplied rank-r separated representation admits r ordinary transforms. We use only this structured statement.

For M158, exact aperiodic correlations can be evaluated by integer convolution, or by a sufficiently padded FFT with certified rounding. Unpadded circular correlations alias the endpoint pairs. No generic Ram change of coordinates removes that issue or guarantees a speedup. The inspected public M158 modules required no mathematical correction in the chains used here. Their equal-curvature case, order of limits, and restricted spectral conclusion are valid details that we retain.

7 Worked certificates and counterexamples

7.1 A three-letter word

Take $a = (1, 1, -1)$. Then $C_1 = 0,\ C_2 = -1$, and

$$\rho_a(t) = 1 - \tfrac{2}{3}\cos(4\pi t), \qquad \|\rho_a\|_\infty = \tfrac{5}{3}, \qquad F(a) = \tfrac{9}{2}.$$

The three-column Gram is

$$T_3(a) = \begin{pmatrix} 1 & 0 & -1/3 \\ 0 & 1 & 0 \\ -1/3 & 0 & 1 \end{pmatrix},$$

with eigenvalues $2/3, 1, 4/3$. The cap-only lower bound $1-2(2/3) = -1/3$ is inconclusive, whereas the exact Gram is well conditioned. For example the rational LDL^* pivots of $T_3 - \frac{1}{2}I$ are $1/2, 1/2, 5/18$, so a physical lower margin $1/2$ is certified without numerical eigenvalue estimation. Here $S_2 = 4$. At $M = 16$, the grid contains a maximum, and Proposition 5.1 gives the conservative upper cap $5/3 + \pi^2/192$.

For the word $a \times a$, of length nine, the only nonzero positive normalized correlations are

$$c_2 = -1/3, \quad c_4 = 1/9, \quad c_6 = -1/3, \quad c_8 = 1/9.$$

Thus $F(a \times a) = 81/40$. Block multiplication need not improve merit factor; the summable-cap hypothesis is doing real work. Repeating this same block indefinitely does not meet that hypothesis.

7.2 An actual zero and a collapsing window margin

For $a = (1, 1)$, $\rho_a = 1 + \cos(2\pi t)$ vanishes at $1/2$. The L-column Toeplitz Gram has diagonal 1 and adjacent entries $1/2$, hence eigenvalues

$$1 + \cos\frac{k\pi}{L+1}, \qquad 1 \le k \le L.$$

Indeed the sine vectors $\sin((j+1)k\pi/(L+1))$, $0 \le j < L$, satisfy the tridiagonal recurrence and the zero endpoint conditions. Every finite Gram is positive definite, but its smallest eigenvalue $1 - \cos(\pi/(L+1))$ tends to zero. Almost-everywhere positivity does not give a uniform infinite-window Riesz lower bound. This word has cap excess one, so it is an illustration of the logical gap, not an asymptotically small-excess construction.

7.3 Normalized products without independent factors

Take any actual list of binary blocks. Equation (4.1) proves the product mass one even though the dilation factors can have overlapping Fourier differences. It is not necessary to label them independent or dissociated. In contrast, the repeated-density example in Remark 3.3 fails precisely this structural identity. Positive tower polynomials $r^{-1/2}\sum_{j=1}^{r} z^{a_j}$, with distinct exponents and $r \geq 2$, have squared value r at $z = 1$. They therefore cannot meet summable cap excesses at infinitely many nontrivial stages. This obstruction is already in TR-item-169 and is not bypassed by the binary-sign product construction.

8 Evidence, advantages, costs, and next questions

8.1 Independently executable checks

The accompanying `M158-exact-checks.py` uses Python integers, rational fractions, and pairs of rational numbers for Gaussian arithmetic. It requires only the standard library. Its final run passed 8,734 parameter cases and 60,174 asserted finite identities or inequalities. One case is a declared word, word pair, matrix window, or parameter choice; one assertion is one explicit check call. Vector equalities may contain several entries and are still counted as one assertion. Parent or cross-review tests are not included.

Exact check group	Cases	Assertions
Binary Parseval, all words up to sign, lengths 1–10	1,023	4,091
Physical Toeplitz energy and positive rational pivots	5,104	30,624
Mixed-radix word identities, lengths 1–5	961	19,524
Sharp cap chords and sublevel inequalities	16	288
Normalized dominated-density tail budgets	399	1,995
Auxiliary rate induction inequalities	501	1,002
Worked words and weighted physical Grams	21	24
Zero-diagonal tail comparison	42	84
Rational cosine enclosures at known angles	42	42
Corrected four-point Fourier butterflies	625	2,500
Total	8,734	60,174

The cosine fixtures use a classical rational enclosure for π as an input. They do not test every angle or constitute a numerical search for flat polynomials. The infinite product theorem, spectral identification, and grid theorem rest on their displayed proofs, not on finite enumeration. No asymptotically flat family was constructed by these tests.

8.2 Actual reading and retained dependencies

The literal source record contains all 2,010 lines in twelve cached text files: 1,757 lines in ten TeX modules, 235 bibliography lines, and 18 README lines. Their unread complements are empty. The complete 584-line private comparison TR-item-169 was read, followed by fresh original proofs in TR248, lines 133–267, 349–419, and 1030–1120, and TR9, lines 139–364 and 450–514. This is 588 original-proof lines, not a reading of both complete original reports. In TR248 the unread complements are 1–132, 268–348, 420–1029, and 1121–1170; in TR9 they are 1–138, 365–449, and 515–779. All other private corpus material remains outside this report's literal-read scope.

The source/proof ledgers distinguish heading maps from actual displays and from checked mathematical deductions. Endpoint hashes are compared before and after reading. Batch snapshot

identity is provenance between two captured Git trees, not live-branch status or proof validation. Public literature was consulted by public title or URL only; no confidential text was submitted online.

The key imported boundaries are explicit: the global sign construction retains its hypergraph and partial-coloring inputs; physical realization retains the classical Morse facts; the public lower-bound and fixed-prefix papers are cited in their stated scopes. No proof of every reference, no entire private-corpus audit, and no validation of a stronger companion theorem is claimed.

8.3 What improves and what it costs

The substantive local increment is the no-floor product theorem and its application to the exact binary word structure. It yields a bounded positive-almost-everywhere density and explicit tail errors, where the source's stated Morse corollary supplies an L^2 density and a nonzero absolutely continuous component. The conditional source-to-system implication is stronger on the coordinate cyclic space, but unchanged on the missing full-space cyclicity gate. The proof uses classical measure theory and is not presented as a historically new infinite-product principle.

The second useful increment is an end-to-end finite certificate: integer autocorrelations determine the physical Gram exactly; rigorous grid enclosures and the second-derivative remainder certify the cap; exact product tails transfer finite margins to the limiting coordinate measure. The trace normalization saves one row-sum term. Original Ram mass and perturbation proofs then give a physically meaningful decoder budget. Generic Gram relabeling or the existing merit inequality alone would not justify the contribution rating.

The advantages are explicit errors, allowance for true zeros, no independence assumption, and exact finite arithmetic. The costs are equally concrete. Summable cap excesses require an infinite selected block sequence; the source gives no effective length threshold. Prefix lengths multiply. The grid constant $S_2(a)$ may be large. Window margins can deteriorate despite positivity of every finite Gram. General weighted banks need mass scaling, and exact rational factorizations have denominator and storage costs. None of these costs disappears by naming a transform.

Three focused next steps are warranted. First, quantify the source's curvature, cutoff, and packing constants sufficiently to turn the block-index error budget into a length-dependent statement. Second, seek sharper window-dependent estimates exploiting autocorrelation cancellation rather than entrywise absolute values. Third, analyze which additional observables are needed to describe the Morse orthogonal complement. That last question is a dynamical realization problem, not an automatic consequence of the bounded density theorem.

9 Conclusion

One-sided near-minimal maxima contain more information than the merit bound alone: with a summable block selection and exact word normalization, they give a positive-almost-everywhere bounded spectral density with quantitative tails. This is a proved local consequence, conditional only at the explicitly named global existence and realization interfaces. Finite Ram Gram and modulus certificates make the retained approximation errors testable without asserting control of every observable or of every polynomial length.

The applicability classification remains L3. The contribution is assessed C3 because of the demonstrated no-floor product strengthening and source-specific finite-to-limit certificate, not because a global Littlewood or Banach conjecture has been independently settled.

References

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[6] Private Ram corpus, *Ram-Calculus Spectral Certificates for Banach's Simple Lebesgue Spectrum Problem*, TR-item-169-astra, September 7, 2026. Local confidential comparison manuscript; complete literal reading.

[7] Private Ram corpus, *Stable Adaptive Ram Frames and Task-Aware Approximation*, TR248. Original local proof source; finite-model, perturbation, physical-coordinate, and moment-certificate passages specified in Section 8.

[8] Private Ram corpus, *FFT Applicability Analysis for the Ram Transform*, Technical Report 2026-04-04-0009. Original local spectral-computation source; the inspected portions and normalization corrections are specified in Sections 6.3 and 8. Uninspected portions are not endorsed.